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Roll No

MI-404 (GS)
B.E. IV Semester Examination, June 2020
Grading System (GS)
Electrical and Electronic Instrumentation
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. Explain the working and principle of enhancement type MOSFET. Also draw its drain and transfer curve.

OR

What are OP-AMP characteristics? Design an adder circuit using OP-AMP to get output expression as under.

$$V_{\text{out}} = -(V_1 + 10 V_2 + 100 V_3)$$

Where V_1 , V_2 and V_3 are inputs. It is given that $R_F = 100 \text{ k}\Omega$.

2. Compare the D.C. Series and Shunt motor on the basis of their operating characteristics.
3. Draw the circuit diagram of a differentiator using OP-AMP and explain its frequency response.
4. Explain the VI characteristics of Zener diode and describe its working as a voltage regulator.
5. Describe the application of dc shunt, series and cumulative compound motors. How do dc motors compete with ac motors? Discuss.
6. Explain briefly DC 2 and 3 wire system.

OR

- a) What do you mean by circuit breaker? Explain its types.
- b) Discuss DC 2 and 3 wire system.
7. Compare:
- i) AC transmission over DC transmission.
- ii) Overhead lines verses Underground cables.

OR

Explain why synchronous motor is not self starting. Also discuss the methods of starting.

8. Answer any two of the following:
- a) Discuss voltage regulation of an alternator using emf or mmf method.
- b) What is magnetizing inrush current and how transformers are protected against this?
- c) Draw the voltage regulator circuit using Zener diode and explain its working.
- d) Write characteristics of an ideal operational amplifier.
